Dongha Lee

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Education

Stanford University — Stanford, CA
M.S. in Electrical Engineering — *Sep. 2026 – Jun. 2028 (Expected)*
- Coursework: Probability and Statistical Inference, Explainable AI with Neuroscience

Chung-Ang University — Seoul, South Korea
B.S. in Electrical and Electronics Engineering — *Mar. 2019 – Feb. 2026*
- GPA: **4.40/4.50** (99.00/100) | *Summa Cum Laude* | Rank: **8/159**

Technical Skills

Programming: Python, C/C++, Bash, MATLAB
Machine Learning: PyTorch, Transformers, PEFT, Accelerate, Computer Vision, LLMs, Multimodal Learning
Systems: CUDA, multi-GPU training (4× NVIDIA RTX 3090), Linux/Ubuntu, Git, SSH, Conda

Research & Engineering Experience

Predicting What Unlearning Erases in Language Models — Stanford, CA
Independent Research | Two manuscripts in preparation — *Oct. 2026 – Present*
- Developed a step-0 gradient kernel predicting per-fact erasure transfer, relearnability, and collateral damage **before model updates**; validated on TOFU up to **7B** ($\rho \approx 0.5$–0.7) and in a replication with hypotheses fixed in advance.
- Identified shared answer-token suppression as a major source of collateral damage; kernel-guided retention reduced worst-case damage by **49–66%**, with adaptive kernel updates improving protection during sequential unlearning.

Computer Vision & Machine Learning (CVML) Lab — Chung-Ang University
Research Assistant | Advisor: Prof. Junseok Kwon — *Nov. 2024 – Aug. 2026*

Activation Boundary Matching for Low-Rank Adaptation — *AAAI 2027, Under Review*
- Developed a task-informed LoRA initialization that transfers activation-boundary information from a short probe adapter, improving early adaptation with **no inference-time overhead**.
- Evaluated across vision and language models up to LLaMA2-7B, outperforming standard LoRA by up to **3.37 points** and ranking first on **4/8 vision evaluations**.

Information-Theoretic Representation Learning (IB/IIB) — *IEIE 2025*
- Analyzed how Information Bottleneck objectives trade off compression against task-relevant feature retention, and how IB-based learning suppresses spurious features under distribution shift.

HI-LoRA — Hybrid Initialization for Low-Rank Adaptation [Thesis] — *Undergraduate Thesis*
- Independently designed a hybrid initialization combining gradient- and activation-based signals; achieved the highest **4-task GLUE average (89.13)** and best or tied-best performance on 3/4 tasks against evaluated PEFT baselines.

ESG Practical Problem-Solving Research Program — Chung-Ang University
Team Leader & Lead Researcher | Advisor: Prof. Jung-Ryun Lee — *May 2024 – Dec. 2024*
- Led a 5-person team building a noise-robust multimodal system combining lip reading and speech recognition.
- Designed a PSNR-based adaptive modality-switching algorithm robust to changing acoustic noise.

Leadership & Creative Experience

Fashion & Commercial Photography [Portfolio] — Grant Studio, Seoul
Studio Assistant | Freelance Photographer — *Mar. 2023 – Jan. 2024*
- Supported productions for clients including **Netflix and Kakao**, handling lighting, equipment, site surveys, and on-set coordination; independently covered the **2023 Australian Fashion Week (AAFW)** as an accredited photographer.

Da Vinci Volunteer Group — Chung-Ang University
President — *Mar. 2019 – Dec. 2020*
- Led outreach and mentoring for **36 underserved students**, directing a virtual mentoring program during COVID-19.

Selected Publications

[1] **Dongha Lee***, J. Park*, M. Kim, and J. Kwon, "Activation Boundary Matching: Task-Informed Initialization for Low-Rank Adaptation." *AAAI 2027, Under Review.* **[arXiv]** **[Code]**

[2] **Dongha Lee** and J. Kwon, "Designing Information Flow in Deep Representations: The Importance of Information Bottleneck and a Case Study in Invariant Learning." *IEIE Summer Conference*, 2025. **[Supplement]**

*Equal contribution